



EDUCATION				
Program	Degree	Institution	% / CGPA	Year
Dual Degree (B.Tech + M.Tech)	Engineering Design, <i>Specialization - Robotics</i>	Indian Institute of Technology Madras	8.3/10	2024
Class XII	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	95.4%	2019
Class X	CBSE	Maharishi Vidya Mandir, Chetput, Chennai	10/10	2017
• Awarded (among 30 out of 800+ students) the IIT Madras Young Research Fellowship based on remarkable research potential. 2021 • Secured 97.6457 Percentile in JEE Advanced and 99.1046 Percentile in JEE Mains out of over 1.5 Million aspirants across India. 2019 • Secured an All India Rank of 51 in UCEED(Undergraduate Common Entrance Examination for Design) conducted by IIT Bombay. 2019				
RELEVANT COURSES AND SKILLS				
Courses	Deep Learning*(CS6910), Introduction to Robotics (ID6040), Field and Service robotics (ED5315), Computation and Visualization (Python, C++ & OpenGL), Control Systems (ED2040)			
MOOCs	Algorithmic Toolbox * (UC San Diego), Intro to tensorflow , Reinforcement Learning (Alberta), Game Dev (Michigan)			
Skills	Languages: Python, C, C++, C#	Tools: VS Code, MATLAB, LaTeX, MS-Office, Unity3D, Fusion360, Linux		
	Frameworks: Tensorflow, Pytorch	Libraries: Pandas, NumPy, Scikit Learn, Matplotlib, OpenCV, PyQt5, OpenGL		
RESEARCH AND PROFESSIONAL EXPERIENCE				
Senai Lab Computer Vision Intern Jun 2022 – Present Proof 2 , Proof 3	<i>Project aims to develop a self-balancing & autonomous bicycle under the guidance of Prof. Raghunathan Rangaswamy.</i> •			
Yaskawa Pvt. Ltd Robotics Intern Jun 2022 – Present Proof 2				
Young Research Fellow Undergraduate Researcher Jun 2021 – Present Proof 2	<i>Research on Path Planning Algorithms for Unmanned Aerial Vehicles under the guidance of Prof. Satadal Ghosh.</i> • Analyze			
Blurgs Computer vision Intern Jun 2021 – Aug 2021	<i>Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action.</i> • Analyzed and tested 7+ State of the art Multiple Object Tracking networks for the startup's first product. • Trained and tested various MOT networks like Siam-MOT and FairMOT for Surveillance with Drones and CCTV. • Used FairMOT to track and get the trail of multiple moving objects and Re-identify objects in real-time.			
Co-curricular Affairs Secretary Avishkar Hyperloop Power Systems Engineer Oct 2020 – Jun 2021	<i>The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country</i> • Winner of the most scalable design award by Zeleros¹ and among the top 5 for mechanical and propulsion. • Designed a batter pack with high discharge rate for higher speeds while ensuring safe power supply to the pod • Worked on a custom-made Battery Management System to maintain voltage, temperature and ensure safety. • Extended battery pack life by 30% by reducing peak operational temperature by 10°C using an Al⁴ heat sink.			
Imagine Pvt. Ltd. Software Dev. Intern Apr 2020 – Aug 2020	<i>Enables collaboration across different locations using VR/AR for purposes such as industrial design, support and sales.</i> • Developed a Unity3D plugin to upload any 3D model into a VR² space using Autodesk Forge APIs³ in real-time. • Built a basic Heroku app to upload and view any 3D model with 60+ file formats in the web browser. • Achieved 100% reduction in staff-hours required for this specific task by automating the entire pipeline.			
PROJECTS				
VR – RL Shooter CVI⁸ Club Co-curricular Affairs Secretary (May 2020 – Jul 2020)	• Built and trained a Multi agent Reinforcement Learning environment where agents play a shooting game. • The environment was made on Unity3D with C#, intractable blocks were included to see exploratory tactics. • Environment is setup to be ported as a Virtual Reality environment where the user can also play the game.			
Path Finder PyQt5⁶ ED3150 (Apr 2021)	• Developed a 2D Path finder from scratch in Python using SOTA⁴ A-Star and Dijkstra Algorithms and PyQt5. • Designed an interactive UI⁵ for visualization using QT Designer⁶; Documented the project for future use.			
CS6910 CS6910 (Apr 2021)	• D.			
CFD⁷ Project (ED4040) (Mar 2021 - Apr 2021)	• Designed an aero foil for the Front wing of a Formula Student car for maximum Downforce/Drag ratio. • Improved Force/Drag ratio by 33% by varying position and angle of the aero foil relative to ground and car. • Performed CFD⁷ analysis to determine maximum possible ratio and the optimal setup for better efficiency.			
Human Powered Segway (ED4060) (Apr 2021 – Jun 2021)	• Ideated a novel design of a fully mechanical version of a Segway under the guidance of Prof. Jayaganthan • Improved efficiency by 67% by replacing conventional circular pedaling system with rectilinear pedaling. • Varied the number of linkages and lengths using MATLAB to optimize the design and improve its potential.			
<i>Gesture Robot</i> ED5080 Project (July 2021- Nov 2021)	• <i>Designed a robotic arm with 2 DOF 8 for identifying and lifting objects using electromagnets.</i> • <i>Implemented proximity sensors for detection and used LED light reflection for sensing color.</i> • <i>Programmed using Arduino mega and added manual and automatic calibration features.</i>			

Product Design ED1011 Project	- Analysed various products and deduced 50+ design changes for maximum customer satisfaction and usability. - Ideated all the necessary parameters and metrics to approach the new design plan for a given product. - Used different product design tools such as Flow method, TRIZ, Product architecture, House of quality etc Analyzed a Roti Maker and deduced 20+ changes to improve customer satisfaction and usability. - Ideated all the necessary parameters and metrics to achieve an ergonomic design plan. - Used different product design tools such as Flow method, TRIZ, Product architecture, House of Quality etc.
Road Junction Safety ED3010 Project	- Modelled a novel road junction design for a hilly region, which can eliminate accidents at a hairpin point. - Designed 15 different innovative junction designs which can reduce road accidents at junctions by 50% - Identified the significant threats in the road infrastructure and suggested feasible alternatives. - Analyzed the design of accident-prone junctions and ideated feasible changes to reduce causality. - Improved road intersections design to bring down the accident rate by 5%.
Gender Inequality ED1034 Project (July 2020- Nov 2020)	- Brainstormed to identify the stereotypical gender inequalities and demonstrated the origin of differences. - Ideated model to ensure workplace equality by seminars, monitoring work division, and other benefits. - Conducted surveys to analyze the response to gender norms in visual storytelling and blogs

POSITION OF RESPONSIBILITIES

Academic Affairs Secretary Placement Core (May 2021 - Present)	Member n
Academic Affairs Secretary Industrial and Public Relations, Coordinator (May 2021 - Present)	- Member of team managing 200+ non-engineering profiles in sectors such as consulting, finance and analytics - Discovered and contacted 20+ potential recruiters based on skill-set requirements and student requests - Responsible for coordinating logistics of 100+ firms and 1000+ final year students in the upcoming season
ShARE, Junior Consultant (May 2021 – Present)	- Selected out of 200 applicants after multiple rounds of Global Interview comprising of consulting case analysis - Working in 'Mobility' Module, undergoing training on focused application of problem-solving in consulting - Part of a global team of 700 students across 15 nations, enabling firms to transition into 'Do Well, Do Good'
Academic Affairs Secretary Internship Team, Coordinator (May 2020- Apr 2021)	- Organized first-ever virtual Internship Day-1 achieving 152 offers from 20 companies across sectors - Part of a team of 25 liaising with companies across multiple sectors for Campus Internships on a rolling basis - Acted as a single Point of Contact for 20+ companies, interacting with 600+ students across departments.
Co-curricular Affairs Secretary CVI ⁸ Club, Coordinator (May 2020- Apr 2021)	- Responsible for the club activities like conducting peer-to-peer sessions, mini-projects, events and projects. - Conducted student run sessions where basics of OpenCV, Machine Learning & Deep Learning were taught.
Co-curricular Affairs Secretary Coding and Logic Events, Coordinator (May 2020- Apr 2021)	- Handled a total footfall of 300+ teams and crafted a total of 50+ logical questions for events in Shaastra'21⁸. - Assisted the team in the successful completion of Hackathons conducted by AstraZeneca and Atlassian Ideated and executed the conduction of 2 brand new events, Mystery Rooms and E-Contest
Co-curricular Affairs Secretary Associate Manager, Startup Services, E-Cell (Nov 2019- Apr 2021)	- Organized Summer Internfair 2020, handling logistics, interviews and congregating around 40+ Startups. - Organized Elevate, a fund-raising competition for startups, bringing in 150+ startups from all over the country - Managed the operational requirements of board members of unicorns throughout the Annual E-Summit¹⁰

EXTRA CURRICULAR ACTIVITIES

Drawing	Won 3 state-level competitions and 5+ District level competitions in drawing (Oil Pastels and Pencils)
Teaching Assistant (GN1101)	
Teaching assistant (GN6106 (HHS))	Happiness Habits and success (course name proof in grade card)
Game Dev. / Gaming	- Love to try out various tools and features in Unity3D; Reached a rank of Silver 2 in Valorant by Riot Games. - Solely developed a complete 3D Game in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.

Sports	• Among the 20 out of 100 students selected for freshman Volleyball training (NSO);
Fusion360 API	• Developed a python script to automatically create models in Fusion360 just by specifying dimensions.

* 8.51 : As of 20 Feb '22; [1] Zeleros – A Spanish company that develops Hyperloop; [2] Virtual Reality; [3] Application Programming Interface; [4] State of the Art ;[5] User Interface;[6] Tool for designing and building graphical user interfaces ; [7] Computational Fluid Dynamics; [8] Computer Vision and Intelligence Club; [9] ISO Certified Annual Technical Festival of IIT Madras; [10] ISO Certified Annual Entrepreneurship Summit; *- ongoing courses